

cards is one reason for this. This bus is 32 bits wide and operates at 33 MHz, and has a maximum data transfer speed of 132 Megabytes per second. This bandwidth is divided between all slots and also the storage devices, like the hard disks.

1.1.14 PCIE (PCI-Express)

PCI-Express is the latest type of expansion slot. Unlike the AGP and PCI buses which transfer data in a parallel fashion, PCIE offers serial data transfer. It is designed to be a point-to-point bus, and so it is not shared by other components. The bus is implemented as a pair, with two lanes per bus. So it can act in full duplex mode, meaning that data can be simultaneously sent and received over different lanes at full speed. It is also modular, in that the number of PCIE lanes can be increased or decreased based on the need of the device. So a PCIE x16 slot that has 16 lanes (8 lanes each for carrying data in one direction) can be used for graphics cards, while a slower PCIE x2 slot can be used for less demanding components like modems and sound cards. Each PCI-Express lane can transfer 2 Gigabits of data per second, which translates to 250 Megabytes per second.

1.1.15 Ports

Every motherboard offers a plethora of ports on its back panel. The most commonly seen ports on today's PCs are PS2, USB, VGA, LAN, Serial, and Audio Sockets.

1.1.16 PS2 port

The PS2 port is used to connect PS2-compatible keyboard and mouse. The violet-coloured port is usually for the keyboard, and the green one is for the mouse.

1.1.17 RAM slots

These refer to the interface to connect RAM modules. Memory is organised in the form of banks, with one or more slots constituting a bank. Each bank has a dedicated channel to the memory controller. Many motherboards, thanks to their Northbridges, implement a dual-channel RAM mode, wherein the effective bandwidth